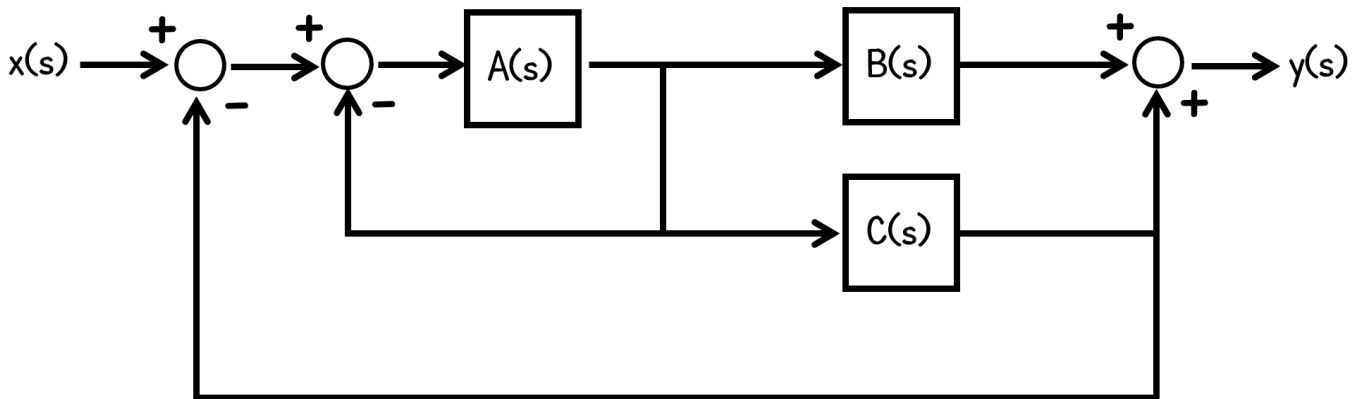


Tutorial 1: Block Diagram Algebra & Modelling

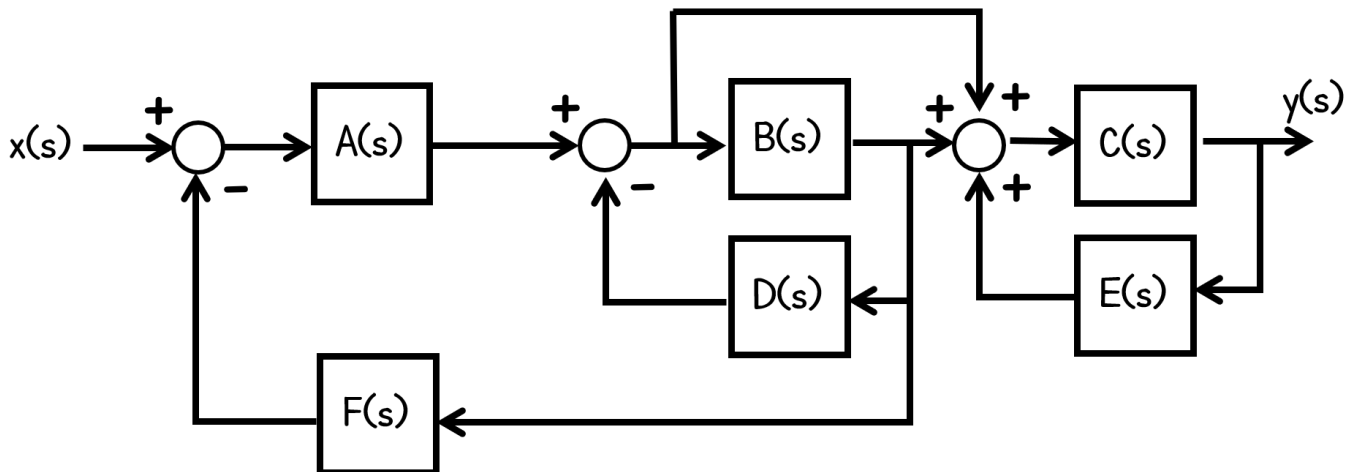
Block Diagrams:

Reduce each of the following block diagrams to one block.

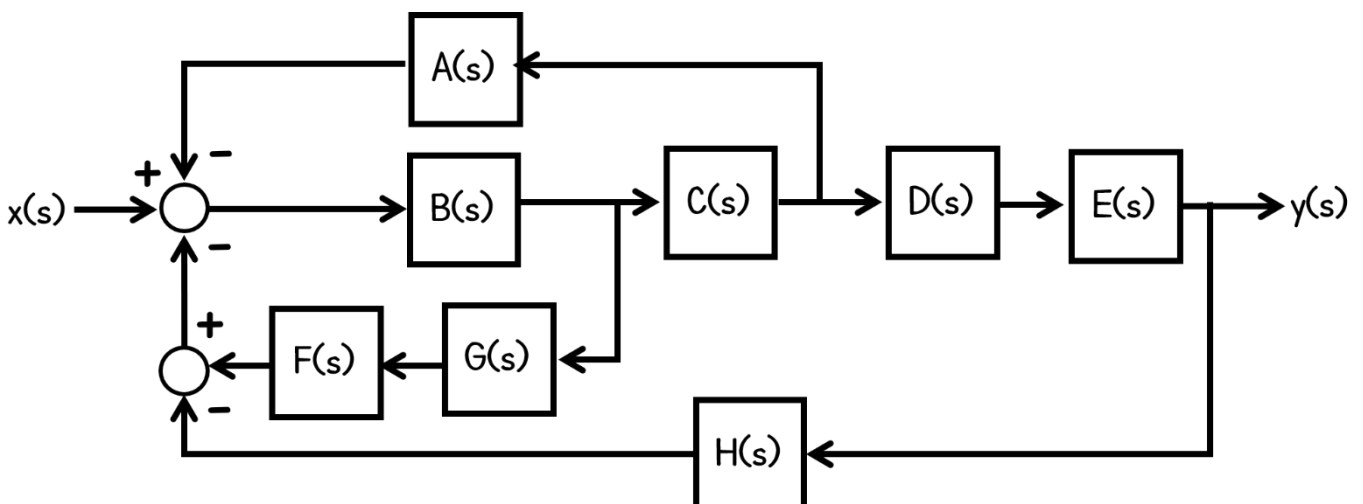
Question 1



Question 2



Question 3



Modelling Questions:

Question 4

During a medical operation an anesthesiologist controls the depth of unconsciousness by controlling the concentration of isoflurane in a vaporized mixture with oxygen and nitrous oxide. The depth of anesthesia is measured by the patient's blood pressure. The anesthesiologist also regulates ventilation, fluid balance, and the administration of other drugs. In order to free the anesthesiologist to devote more time to the latter tasks, and in the interest of the patient's safety, we wish to automate the depth of anesthesia by automating the control of isoflurane concentration. Draw a functional block diagram of the system showing pertinent signals and subsystems (Meier,1992)

Question 5

Neuromechanics is an area of biological research that studies how the nervous system controls the musculoskeletal system to bring about movement. A neuromechanics problem that the African Robotics Unit is interested in investigating is the role of the cheetah's tail in balancing its body. Gavin and I previously travelled up to the Ann van Dyk Cheetah Centre to film cheetahs being transported by bakkie. As the car bumps over the rough dirt roads around the sanctuary, the animals often lose their footing and are seen swinging their tails to recover.

Your task is to draw a block diagram modelling the cheetah in the vehicle.

Here is the information your diagram must capture:

- The cheetah can be broken down into the following subsystems: the nervous system $N(s)$, the legs $L(s)$, the tail $T(s)$, the body $B(s)$, the eyes $E(s)$, the vestibular system $V(s)$ and the proprioceptive system $P(s)$.
- The cheetah is trying to control its body angle $\theta(s)$. It uses a combination of the eyes, vestibular system and proprioceptive system to detect this angle. You may consider the sensed angle to be the average of these three readings.
- The nervous system considers the difference between the desired angle $r(s)$ and the sensed position, and sends a nervous impulse $i(t)$ to the legs and tail to command them to respond.
- The legs and tail respond to the impulse, producing a combined torque $\tau(s)$ that acts to correct the body angle.
- The motion of the vehicle $c(s)$ also affects the angle of the body